

Solutions

1. **Solution:** Because a point on a line makes its equation true. The intersection point lies on *both* lines, so it satisfies both equations simultaneously; hence it solves the system.
2. **Solution:** True. Rearranging $2x - y = 3$ gives $y = 2x - 3$. Any point on $y = 2x - 3$ makes $2x - y = 3$ true.
3. **Solution:** From $(0, 5) \rightarrow (1, 7)$, slope $m_1 = 2$, so $\ell_1 : y = 2x + 5$. Line ℓ_2 has slope 2 and passes $(0, -1)$: $y = 2x - 1$. Parallel, different b : no intersection $\Rightarrow$ no solution.
4. **Solution:** Line: $y = -3x + 4$. Test $(2, -1)$: $\text{RHS} = -3(2) + 4 = -2$. Compare -1 vs. -2 : since $-1 > -2$, point is **above** the line.
5. **Solution:** From $6x - 2y = 10 \Rightarrow -2y = -6x + 10 \Rightarrow y = 3x - 5$. With $y = 3x + 1$, slopes equal, different intercepts $\Rightarrow$ parallel, **no solution**.
6. **Solution:** (a) Pass: $C_p(x) = 25 + 2x$. Pay-go: $C_g(x) = 5x$. (b) $25 + 2x = 5x \Rightarrow x = 25/3 \approx 8.33$ rides. (c) Pass cheaper for $x > \frac{25}{3}$; pay-go cheaper for smaller x .
7. **Solution:** (a) Plug into $y \geq -\frac{1}{2}x + 4$: at $(4, 1)$, $\text{RHS} = 2$, and $1 \geq 2$ is false. (b) $(0, 4)$: $\text{RHS} = 4$, $4 \geq 4$ true. (c) $\geq$ means boundary **solid**.
8. **Solution:** Example: $(2, 2)$. Check $2 < 2(2) - 1 = 3$ true; and $2 \geq -2 + 3 = 1$ true.

9. **Solution:** $m = \frac{5-(-1)}{7-3} = \frac{6}{4} = \frac{3}{2}$. Using $(3, -1)$: $-1 = \frac{3}{2}(3) + b \Rightarrow -1 = 4.5 + b \Rightarrow b = -5.5$. So $y = \frac{3}{2}x - 5.5$. Test $(0, 0)$: $0 > -5.5$ true $\Rightarrow$ included.

10. **Solution:** Same slope m , different intercepts $b \neq c \Rightarrow$ parallel distinct lines $\Rightarrow$ **no solutions**.

11. **Solution:** Any two nonparallel lines through $(2, -3)$. Example: $y = -x - 1$ and $y = \frac{1}{2}x - 4$. Both give -3 when $x = 2$.

12. **Solution:** Between them means $2x - 5 \leq y \leq 2x + 1$ (either order is fine; both $\leq / \geq$ with same slope).

13. **Solution:** $y > 3x - 2$ is dashed; $y \geq -x + 6$ is solid.

14. **Solution:** Points $(0, -2)$ and $(4, 0)$ give slope $m = \frac{0-(-2)}{4-0} = \frac{2}{4} = \frac{1}{2}$. Line: $y = \frac{1}{2}x - 2$. For $(3, 1)$: $\text{RHS} = \frac{1}{2} \cdot 3 - 2 = -0.5$; since $1 > -0.5$, $(3, 1)$ is **above**.

15. **Solution:** L_1 : slope -1 through $(2, 3)$: $y = -x + 5$. L_2 through $(2, 3), (6, 7)$ has slope $m = 1$, so $y = x + 1$. They both pass through $(2, 3)$, hence intersection is $(2, 3)$.

16. **Solution:** (a) Weight: $x + y = 10$. Cost: $8x + 4y = 6 \cdot 10 = 60$. (b) The blend must satisfy *both* constraints; graphically the unique mass split that meets both lines is their intersection.

17. **Solution:** Take test point in Q3 like $(-1, -1)$. For (i): $-1 \leq 2(-1) + 5 = 3$ true, so third quadrant points *can* satisfy (i). For (ii) $-1 \geq 3$ false. So choose **(i)**.

18. **Solution:** Example: $y > 2x + 1$ works: $(1, 4)$ gives $4 > 3$ true; $(1, 0)$ gives $0 > 3$ false. Many answers possible.

19. **Solution:** Slope $-\frac{3}{2}$, intercept 2: base line $y = -\frac{3}{2}x + 2$. “At or above” $\Rightarrow y \geq -\frac{3}{2}x + 2$.

20. **Solution:** Rewrite $4x - 2y = -14 \Rightarrow -2y = -4x - 14 \Rightarrow y = 2x + 7$. Both equations define the same line $\Rightarrow$ **infinitely many** solutions (all points on the line).

21. **Solution:** (a) $y < \frac{3}{4}x + 2$ is **dashed**. $y \geq -2x - 1$ is **solid**. (b) Example point $(0, 0)$: $0 < 2$ true and $0 \geq -1$ true; so $(0, 0)$ is in the intersection.

22. **Solution:** Example $(x, y) = (1, 2)$: check $2 \leq 1 + 1$ true and $2 > -2 + 4 = 2$ false (strict!). Try $(2, 2)$: $2 \leq 3$ true and $2 > 0$ true. So $(2, 2)$ works.

23. **Solution: No solution**, because parallel distinct lines never meet; no point can satisfy both equations.

24. **Solution:** Example (a square):

$$\begin{cases} -1 \leq x \leq 2 \\ 1 \leq y \leq 4 \end{cases}$$

As inequalities: $x \geq -1$, $x \leq 2$, $y \geq 1$, $y \leq 4$ (any two non-parallel pairs suffice). A lattice point inside: $(0, 2)$.